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BATCH-	16 B.Tech C.S.E

Programming In C-HDOC DAY_10

Problem_1

Write a C program to input time in seconds and convert it into days, hours, minutes, and seconds format.

ALGORITHM

1. Start.
2. Declare 4 variables as integers-sec days min and hour.
3. Input time in seconds and save it as sec.
 - 3.1 Calculate minutes using $\text{sec}/60$.
 - 3.2 Calculate hours using $\text{min}/60$.
 - 3.3 Calculate days using $\text{hours}/24$.
4. Display time using statement-

```
printf("time is %d days %d hours %d minutes %d seconds",days,hour-(days*24),min-(hour*60),sec-(min*60));
```
5. End.

PSEUDOCODE

BEGIN

DECLARE sec, min, hour, days AS INTEGER

INPUT sec

$\text{min} \leftarrow \text{sec} / 60$

$\text{hour} \leftarrow \text{min} / 60$

$\text{days} \leftarrow \text{hour} / 24$

DISPLAY "Time is", days, "days",

hour - (days × 24), "hours",

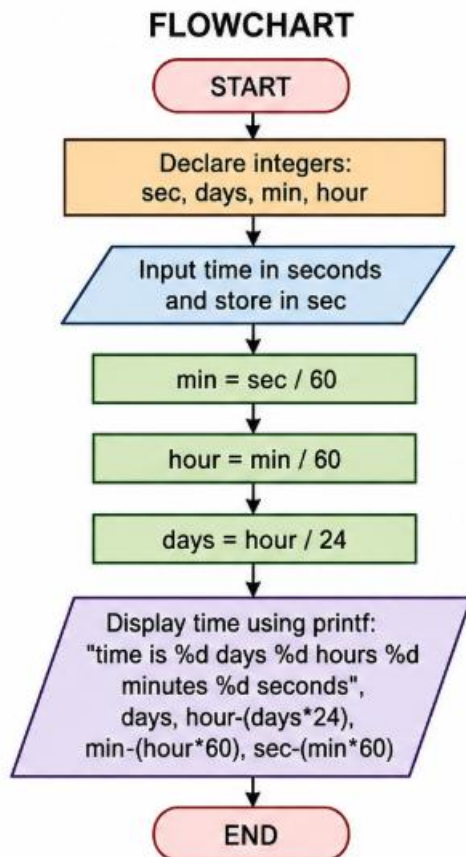
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min - (hour × 60), "minutes",

sec - (min × 60), "seconds"

END

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C PROGRAM CODE

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int sec,min,hour,days;
```

```
    printf("enter time in seconds\n");
```

```
    scanf("%d",&sec);
```

```
    min=sec/60;
```

```
    hour=min/60;
```

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```
days=hour/24;

printf("time is %d days %d hours %d minutes %d seconds",days,hour-(days*24),min-
(hour*60),sec-(min*60));

return 0;

}
```

Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	96603	time is 1 days 2 hours 50 minutes 3 seconds	time is 1 days 2 hours 50 minutes 3 seconds	PASS
2	112296	time is 1 days 7 hours 11 minutes 36 seconds	time is 1 days 7 hours 11 minutes 36 seconds	PASS

Problem_2

Write a program to input a year and check whether it is a leap year or not.

ALGORITHM

1. Start.
2. Input year in AD as variable year.
3. Check if year is divisible by 4 if yes then proceed to step3 otherwise print not a leap year.
 - 3.1_if year is divisible by 100 then proceed to step3.2 ,otherwise display the messege a leap year.
 - 3.2_If year is divisible by 400 then display the messege a leap year,else display not a leap year.
4. End.

PSEUDOCODE

START

INPUT year

IF year is divisible by 4 THEN

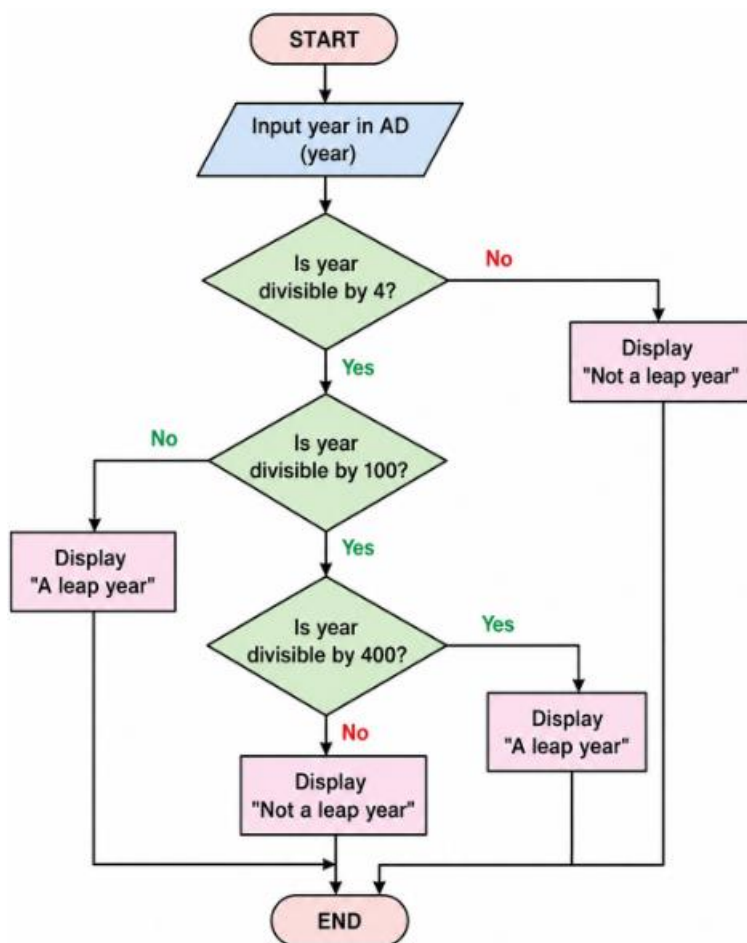
IF year is divisible by 100 THEN

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```
IF year is divisible by 400 THEN
    DISPLAY "A leap year"
ELSE
    DISPLAY "Not a leap year"
END IF
ELSE
    DISPLAY "A leap year"
END IF
ELSE
    DISPLAY "Not a leap year"
END IF

END
```

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C PROGRAM CODE

```
#include <stdio.h>

int main()
{
    int year;
    printf("enter the year\n");
    scanf("%d",&year);
    if(year%4==0)
    {
        if(year%100==0)
        {
            if(year%400==0)
            {
                printf("leap year\n");
            }
            else
            {
                printf("not a leap year\n");
            }
        }
        else
        {
            printf("leap year\n");
        }
    }
    else
    {
        printf("not a leap year\n");
    }
    return 0;
}
```

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Test Case	Input	Expected Output	Actual Output	Result (Pass/Fail)
1	1000	not a leap year	not a leap year	PASS
2	2024	a leap year	a leap year	PASS